

**UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2026 INVITATIONAL A**

Biology

B01. E
B02. D
B03. C
B04. B
B05. D
B06. C
B07. A
B08. A
B09. D
B10. E
B11. C
B12. A
B13. D
B14. B
B15. E
B16. B
B17. B
B18. D
B19. C
B20. C

Chemistry

C01. D
C02. B
C03. D
C04. C
C05. A
C06. C
C07. E
C08. A
C09. B
C10. E
C11. D
C12. D
C13. C
C14. B
C15. B
C16. A
C17. C
C18. A
C19. E
C20. A

Physics

P01. D
P02. A
P03. E
P04. B
P05. C
P06. C
P07. A
P08. B
P09. D
P10. D
P11. E
P12. A
P13. C
P14. A
P15. D
P16. B
P17. B
P18. E
P19. E
P20. C

CHEMISTRY SOLUTIONS – UIL INVITATIONAL A 2026

- C01. (D) The nitrate ion is NO_3^- , so there are four atoms.
- C02. (B) This is an acid-base reaction: $\text{Al}(\text{OH})_3(s) + 3 \text{HCl}(aq) \rightarrow \text{AlCl}_3(aq) + 3 \text{H}_2\text{O}(\ell)$
- C03. (D)
- C04. (C) A single bond and the first bond of a multiple bond are sigma bonds. The second bond in a double bond and the second and third bond in a triple bond are pi bonds.
- C05. (A) $n = PV/RT = (1)(14.0)/(0.08206)(298) = 0.5725 \text{ mol}$.
 $0.5725 \text{ mol} \times 4.00 \text{ g/mol} = 2.29 \text{ g}$
- C06. (C) Tungsten is the only one of the five metals whose density is greater than mercury's.
- C07. (E) $\text{HCl}(aq) + \text{NaHCO}_3(aq) \rightarrow \text{NaCl}(aq) + \text{H}_2\text{O}(\ell) + \text{CO}_2(g)$
- C08. (A)
- C09. (B)
- C10. (E) At equilibrium the forward and reverse reactions continue, but at equal rates, so there is no net change in the amounts present. The actual ratio of reactants to products at equilibrium will vary depending on the reaction.
- C11. (D) For any aqueous equilibrium, the equilibrium constant is equal to the concentrations of the products divided by the concentrations of the reactants (leaving out anything that is solid or liquid). Since the dissociation equation for a weak acid is $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, the products are H^+ and A^- and the reactant is HA .
- C12. (D)
- C13. (C) The rate law expresses the rate of the reaction (typically units like M/sec) based on the concentrations of the reactants at any point in the reaction. The general form of the rate law for a reaction such as $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$ is $\text{rate} = [\text{A}]^x[\text{B}]^y$, where x and y are usually 0, 1, or 2.
- C14. (B) Argon is the only monatomic gas among the answer choices. The other gases are all diatomic.
- C15. (B) Moles $\text{OH}^- = 0.230 \times 0.025 = 0.00575$. moles $\text{H}^+ = 0.120 \times 0.050 = 0.00600 \text{ mol}$.
Net 0.00025 mol H^+ in a total volume of $0.350 \text{ L} = 0.0007143 \text{ M H}^+$.
 $\text{pH} = -\log(0.0007143) = 3.15$
- C16. (A)
- C17. (C) A molecule reacting with O_2 to produce CO_2 and H_2O is a combustion reaction.
- C18. (A) The compound with the highest percent Cu by mass is the one that has the lowest mass of other stuff in it. SO_4 and NO_3 obviously have more mass than O, and so do Cl and S, so CuO has the highest percent copper by mass.
- C19. (E) Barium bromide is BaBr_2 and barium bromite is $\text{Ba}(\text{BrO}_2)_2$.
- C20. (A) The Faraday constant $\mathcal{F}$ (see the data sheet) is the total charge on one mole of electrons and is equal to $96,485 \text{ C/mol}$. To reduce 0.200 mol of Ag^+ ions to solid Ag you need 0.200 mol of electrons. $96,485 \text{ C/mol} \times 0.200 \text{ mol} = 19,297 \text{ C}$.

PHYSICS SOLUTIONS – UIL INVITATIONAL A 2026

- P01. (D) page 10: "...just at the end of 1895.... Wilhelm Röntgen, professor of physics at Würzburg University in Bavaria, had found a new type of invisible radiation which could penetrate solid, opaque objects! Röntgen's rays came from vacuum tubes designed to study cathode rays. He called this radiation 'X' because nothing was known about it."
- P02. (A) page 21: "Marie then began a study of the magnetic properties of different kinds of steel for the Society for the Encouragement of National Industry."
- P03. (E) page 24: "Some years earlier Pierre Curie and his brother Jacques had found that quartz crystals gave off electrical signals when they were compressed. Pierre used this property (known as the *piezo-electric effect*) to devise an unusually sensitive electroscope...."
- P04. (B) When plotted on a Hertzsprung-Russell diagram (a plot of luminosity versus temperature) stars show several distinct groupings. Groups of giant and supergiant stars, as well as protostars and white dwarves can be identified. Most stars, however, fall along a diagonal line from the upper left to the lower right of the diagram. These are the main sequence stars. The main sequence represents where a star will spend most of its lifetime, and the time during which it is powered by hydrogen fusion in its core.
- P05. (C) Correct order of operation requires that we do the subtraction in the numerator first. For subtraction (and addition) it is the decimal place that matters: the answer will only be significant to the same place as the input number with the least decimal places. The number 11.31 is significant to the hundredths place, while the number 6.403 is significant to the thousandths place. The least of these is the hundredths place, so the result of the subtraction will be significant only to the hundredths place:
 $11.31 - 6.403 = 4.90[7]$. The "7" in this case is not significant, but we keep it because we will not round until the entire calculation is finished.
For the division, the result will have the same total number of significant figures as the input number with the least amount of significant figures. The numerator 4.90[7] has three significant figures. The denominator 1.2188 has five significant figures. So, the result will have the least of these – three significant figures. Thus, $X = \frac{4.90[7]}{1.2188} = 4.02[6091] \approx 4.03$.
- P06. (C) The kinematic equation to use is $v_f = v_i + at$. The squirrel starts from rest, so $v_i = 0$. Putting in the final velocity and the acceleration gives: $11.4 = 0 + (6.65)t \rightarrow t = 1.71s$.
- P07. (A) The diagram given also represents the horizontal component of the free-body force diagram, so we have what we need to solve the problem. The two forces act in opposite directions, so the net horizontal force acting on the box is the difference of the two forces: $\sum F_x = 18.0 - 13.0 = 5.0 N$ (to the right, in this case). Now we use Newton's second law to find the acceleration: $\sum F_x = ma \rightarrow 5.0N = (5.75kg)a \rightarrow a = 0.870m/s^2$. Note: there is no motion in the vertical direction, so we do not need to consider any vertical forces.
- P08. (B) Initially, we have elastic potential energy stored in the compressed spring. Once released, this elastic potential energy converts into the kinetic energy of the rock. There is also a little gravitational potential energy resulting from the lifting of the rock as the spring uncompresses. Thus, we have:
 $EPE_i = KE_f + GPE_f \rightarrow \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + mgx \rightarrow$
 $(240)(0.055)^2 = (0.042)v_f^2 + (0.012)(9.80)(0.055)$. This gives a launch velocity of the rock of
 $0.726 = 0.042v_f^2 + 0.0065 \rightarrow v_f = 4.03m/s$.

- P09. (D) We only need to consider rotational equilibrium to solve the problem. Let's start with the torques acting on the meter stick. Torques are produced by both the known and unknown masses. The normal force at the fulcrum does not provide a torque since it acts at the axis of rotation and has a torque arm of zero. The known mass produces a torque oriented in the counterclockwise direction, while the unknown mass produces a clockwise-oriented torque. Thus, we have the rotational equilibrium equation:
 $\sum \tau = 0 \rightarrow m_k g r_k \sin \theta_k - m_u g r_u \sin \theta_u = 0$. The gravitational forces are in the vertical direction (downward), and the meter stick is horizontal, so both angles are $\theta_k = \theta_u = 90.0^\circ$. The fulcrum serves as the axis of rotation, so all torque arm distances will be measured from the 50.0cm mark. For the unknown mass, the torque arm is $r_u = 80.0 - 50.0 = 30.0\text{cm}$. For the known mass, the torque arm is $r_k = 50.0 - 10.7 = 39.3\text{cm}$. Putting in the known values gives:
 $(0.130)(9.80)(0.393) \sin(90.0) - m_u(9.80)(0.300) \sin(90.0) = 0$. This leads to $0.501 = 2.94m_u$ and an unknown mass of $m_u = 0.170\text{kg} = 170\text{g}$.
- P10. (D) The equation for the period of a pendulum is $T = 2\pi \sqrt{\frac{L}{g}}$. We do not know the length of the pendulum, but we can find that using the information given. On Earth: $T = 1.50\text{s} = 2\pi \sqrt{\frac{L}{(9.80)}} \rightarrow$
 $L = (9.80) \left(\frac{1.50}{2\pi} \right)^2 = 0.5585\text{m}$. Now, on the Moon, the length of the pendulum is the same, but we have a different gravitational acceleration, so we get a period of $T = 2\pi \sqrt{\frac{L}{g_m}} = 2\pi \sqrt{\frac{0.5585}{1.62}} = 3.70\text{s}$.
- P11. (E) We have the mass of the stone, so we only need to determine its volume. The volume of the stone equals the amount that the water level rose when the stone was placed in the graduated cylinder. That is $V_{\text{stone}} = 96.0\text{ml} - 60.0\text{ml} = 36.0\text{ml}$. The density is then
 $\rho = \frac{m_{\text{stone}}}{V_{\text{stone}}} = \frac{117.0\text{g}}{36.0\text{ml}} = \frac{0.1170\text{kg}}{0.0000360\text{m}^3} = 3250\text{kg/m}^3$.
- P12. (A) First, we combine the two resistors that are in parallel: $\frac{1}{R_p} = \frac{1}{250} + \frac{1}{450} \rightarrow R_p = 160.7\Omega$. The parallel group is in series with the remaining resistor. Combining these gives the total resistance of the circuit: $R_T = 300 + 160.7 = 460.7\Omega$. Now we can find the current produced by the power source: $V_T = I_T R_T \rightarrow 12.0 = I_T(460.7) \rightarrow I_T = 0.0260\text{A}$. Because they are in series, this same current flows through both the 300Ω resistor and the 160.7Ω parallel group ($I_{300} = I_p = I_T = 0.260\text{A}$). Focusing on the parallel group, we find the voltage across the parallel group: $V_p = I_p R_p = (0.260)(160.7) = 4.19\text{V}$. Because the resistors are in parallel, this is the same voltage across the 250Ω and 450Ω resistors ($V_{250} = V_{450} = V_p = 4.19\text{V}$). Now we can find the current flowing through the 450Ω resistor: $V_{450} = I_{450} R_{450} \rightarrow 4.19 = I_{450}(450) \rightarrow I_{450} = 0.0093\text{A} = 9.3\text{mA}$.
- P13. (C) The force of one charge on another is given by Coulomb's Law. Since we aren't concerned about direction, we can use the equation without worrying about the vector: $|F| = \frac{kQ_1Q_2}{r^2}$. The distance between the charges is $r = 25\text{cm} - (-40\text{cm}) = 65\text{cm} = 0.65\text{m}$. So, the force on the $17.0\mu\text{C}$ charge due to the $12.0\mu\text{C}$ charge is $|F| = \frac{(8.99 \times 10^9)(12 \times 10^{-6})(17 \times 10^{-6})}{(0.65)^2} = 4.34\text{N}$.
- P14. (A) The force on a current-carrying wire due to a magnetic field is given by $F = ILB \sin \theta$. Since the magnetic field is oriented perpendicular to the wire, we have $\theta = 90.0^\circ$. We also convert the magnetic field strength into Teslas: $B = 400.0\text{G} = 0.04000\text{T}$. Plugging it all in, we obtain a force of $F = ILB \sin \theta = (50.0)(0.280)(0.04000) \sin(90.0) = 0.56\text{N}$.

- P15. (D) The angle given is relative to the surface of the water, but we need to find the angle relative to the normal line to the surface. That is, the incident angle is $\theta_i = 90.0 - 64.0 = 26.0^\circ$. It is this angle which we use with Snell's Law. Since the light beam starts in the water, the initial index of refraction is 1.33, and the final index of refraction is 1.00 (air). Using Snell's Law, we get $n_i \sin \theta_i = n_r \sin \theta_r \rightarrow (1.33) \sin(26.0) = (1.00) \sin \theta_r \rightarrow 0.583 = \sin \theta_r$. So, the refracted angle is $\theta_r = 35.7^\circ$. This is the angle relative to the normal line to the surface. Thus, the angle of the beam in air relative to the surface of the water is $\theta = 90.0 - 35.7 = 54.3^\circ$.
- P16. (B) The mirror is convex, so the focal length is negative. We calculate a focal length for the mirror of $f = (-)\frac{R}{2} = -\frac{72.0}{2} = -36.0\text{cm}$. The object (the moth) is located at $p = 22.0\text{cm}$. Thus, we can find the image location by using $\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \rightarrow \frac{1}{22.0} + \frac{1}{q} = \frac{1}{-36.0} \rightarrow q = -13.7\text{cm}$. Since the image location is negative, this confirms that the image is virtual (as stated), and the magnitude of the distance of the virtual image from the mirror's surface is $|q| = 13.7\text{cm}$.
- P17. (B) The uncertainty principle for position and momentum is given by $\Delta x \Delta p \geq \frac{\hbar}{2}$. As usual, we will use the equality to estimate the uncertainty. Putting in the given uncertainty in position, and inserting the value for $\hbar$, we obtain: $(25.0 \times 10^{-9})\Delta p = \frac{1.054 \times 10^{-34}}{2} \rightarrow \Delta p = 2.11 \times 10^{-27}\text{kgm/s}$. This is the uncertainty in momentum. The electron is trapped, and this small uncertainty in momentum is non-relativistic, so we can relate the uncertainty in momentum to the uncertainty in velocity using classical equations: $\Delta p = m\Delta v \rightarrow 2.11 \times 10^{-27} = (9.11 \times 10^{-31})\Delta v$. This gives an uncertainty in velocity of $\Delta v = 2.3 \times 10^3 \approx 2 \times 10^3\text{ m/s}$. Note: You may find variations in the equation for the uncertainty principle, but all of them will lead to an order of magnitude of $\Delta v \approx 10^3$.
- P18. (E) Since this particle is a meson, it must consist of a quark and an anti-quark. This eliminates choices A and B (which are baryons). The strangeness must be -1 , so the quark must be an s-quark. This eliminates choices D and F. The charm of -1 tells us that the anti-quark must be an anti-c-quark. Thus, the correct answer is (anti-c)(s), or $\bar{c}s$, which is choice E. This also gives the correct charge number of -1 .
- P19. (E) On a plot of position and time, the velocity is related to the slope of the line. In this case, the slope is the inverse of the velocity, which can be seen from the units of the slope. To begin, let us calculate the slope itself (with units). We first choose two points on the best fit line that are not data points. I shall choose (24m, 1.0s) and (73m, 3.0s). These points give a slope of $\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3.0\text{s} - 1.0\text{s}}{73\text{m} - 24\text{m}} = \frac{2\text{s}}{49\text{m}} = 0.041\text{s/m}$. The velocity is then $v = \frac{1}{\text{slope}} = \frac{1}{0.041\text{s/m}} = 24.5\text{m/s} \approx 25\text{m/s}$.
- P20. (C) Once again, it will be the slope of the line that is important. The electric potential near a point charge is described by the equation $V = \frac{kQ}{r}$. Relating this to our plot shows: $V = (kQ)\left(\frac{1}{r}\right)$. Thus, we see that the slope is equal to kQ . To calculate the slope, we choose two points on the best fit line that are not data points. I shall choose (1.1V, 10m^{-1}) and (3.5V, 30m^{-1}). These points give a slope of $\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{30\text{kV} - 10\text{kV}}{3.5\text{m}^{-1} - 1.1\text{m}^{-1}} = \frac{20000\text{V}}{2.4\text{m}^{-1}} = 8330\text{Vm}$. Setting this equal to the quantity that we derived from the equation gives: $kQ = 8330\text{Vm} \rightarrow \left(8.99 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2}\right)Q = 8330\text{Vm}$. This gives $Q = 9.3 \times 10^{-7} \frac{\text{C}^2\text{V}}{\text{Nm}} = 9.3 \times 10^{-7}\text{ C} = 930\text{nC} \approx 950\text{nC}$.